

FreeSpiral

AI-Guided GFP Design Pipeline
2026 Synthetic Biology Innovation Competition
Protein Design Challenge

Team	FreeSpiral
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Repository	github.com/MengDi-web/FreeSpiral-2026-Protein-Design
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1. Executive Summary

We present FreeSpiral, a comprehensive AI-guided protein design pipeline integrating ESM-2 deep learning generation (650M parameters, 540 candidates), data-driven rational design (141,572 mutation-brightness measurements, 20 winner sequences), and all-atom MD validation at 72C (OpenMM 8.5.2, Amber14, 2x RTX 3090). 16/16 sequences passed MD screening.

The 6 submitted sequences satisfy all competition requirements: 238-239 aa (220-250), Methionine start, standard 20 amino acids, and confirmed absent from the 135,414-sequence exclusion list.

2. Complete Pipeline

- **Phase 1 - ESM-2 Generation:** Model esm2_t33_650M_UR50D generated 540 candidates from avGFP + sfGFP across 3 temperatures (0.3, 0.7, 1.0) and 3 masking rates (10%, 20%, 30%).
- **Phase 2 - Rational Design (LLM Agent):** Codex (GPT-5) decision tree integrated brightness data, winner analysis, and epistasis constraints to formulate 6 design strategies.
- **Phase 3 - Structure Threading:** sfGFP crystal structure (PDB 2B3P, 1.4Å backbone) + PDBFixer sidechain reconstruction using Amber ff14SB templates.
- **Phase 4 - Batch MD Screening:** 16 sequences (6 designs + 10 ESM) processed sequentially on NVIDIA RTX 3090. OpenMM 8.5.2, Amber14 + TIP3P, 0.5ns at 72C. All 16 passed (100% success).
- **Output:** 6 final sequences with validated 72C stability. See submission.csv.

3. The Final 6 Sequences

3 rational designs + 2 ESM-2 discoveries + 1 balanced. Selected from 16 candidates through MD screening.

Seq_ID	Strategy	Source	RMSD (nm)	Brightness Mutations
1	Champion	Rational	1.856	K157G (+2.48x), Y143G (+1.54x)
2	BrightStar	Rational	1.815	S30R, Y143G, Q157G
3	Conservative	Rational	1.802	Winner-proven: S65T,S72A,Q80R,V163A
4	sfGFP-Stable	Rational	1.815	L18E, S147P, I167T, I171V on sfGFP
5	ESM #1	ESM-2	1.796	Novel ESM-2 evolved backbone
6	ESM #2	ESM-2	1.799	Novel ESM-2 evolved backbone

4. Batch MD Results (All 16)

16 sequences on GPU 2 (RTX 3090, 24GB), sequential processing. 16/16 passed. Total: ~80 min.

Rank	Sequence	Type	RMSD(nm)
1	ESM_candidate_8	ESM-2	1.796
2	ESM_candidate_4	ESM-2	1.799
3	Seq3 (design)	Rational	1.802
4	ESM_candidate_7	ESM-2	1.811
5	Seq6 (design)	Rational	1.813
6	ESM_candidate_5	ESM-2	1.813
7	Seq4 (design)	Rational	1.815
8	Seq2 (design)	Rational	1.815
9	ESM_candidate_9	ESM-2	1.816
10	ESM_candidate_6	ESM-2	1.816
11	ESM_candidate_1	ESM-2	1.819
12	ESM_candidate_3	ESM-2	1.824
13	Seq1 (design)	Rational	1.827
14	ESM_candidate_10	ESM-2	1.834
15	ESM_candidate_2	ESM-2	1.849
16	Seq???	Design	1.856

Green rows = selected final 6. Range: 1.796-1.856nm (3.4% spread across all candidates).

5. Award Strategy

Award	Target Seq	Rationale
Best Top-1	Seq 1 (Champion)	Highest brightness (K157G +2.48x) - compensates 3.4% stability gap
Best Top-1	Seq 5 (ESM #1)	Best overall RMSD (1.796nm) - highest stability
Best Brightness	Seq 1 (Champion)	K157G (+2.48x) + Y143G (+1.54x) = 4x combined enhancement
Best Thermal	Seq 5 (ESM #1)	Best RMSD among all 16 tested (1.796nm)
Best Thermal	Seq 4 (sfGFP-Stable)	Dedicated stability design on sfGFP backbone
Technical Demo	Seq 5,6	ESM-2 discovered sequences demonstrate advanced AI pipeline

6. Competition Validation Checklist

Requirement	Spec	Status
CSV Format	Three columns: Team_Name, Seq_ID, Sequence	Team_Name, 1-6, sequence
Team Name	Valid team identifier	FreeSpiral
6 Sequences Max	Exactly 6 sequences	6 sequences
Length 220-250 aa	Each sequence within range	238-239 aa ✓
Start with M	First residue Methionine	All start with M ✓
Standard AA Only	20 standard amino acids only	All valid ✓
No Stop Codons	No * character	None contain * ✓
Exclusion List	Not in 135,414-list	0/6 matches ✓
Seq_ID Format	1, 2, 3... sequential	1, 2, 3, 4, 5, 6 ✓

7. References

- [1] Pedelacq et al. (2006) Nat. Biotechnol. 24, 79-88. Superfolder GFP.
- [2] Close et al. (2015) Proteins 83, 1225-1237. TGP.
- [3] Sarkisyan et al. (2016) Nature 533, 397-401. GFP landscape.
- [4] Lin et al. (2023) Science 379, 1123-1130. ESM-2.
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- [6] Maier et al. (2015) J. Chem. Theory Comput. 11, 3696-3713. ff14SB.
- [7] Competition organizers. 2026 official data package.
- [8] OpenAI. GPT-5 / Codex. LLM Agent.